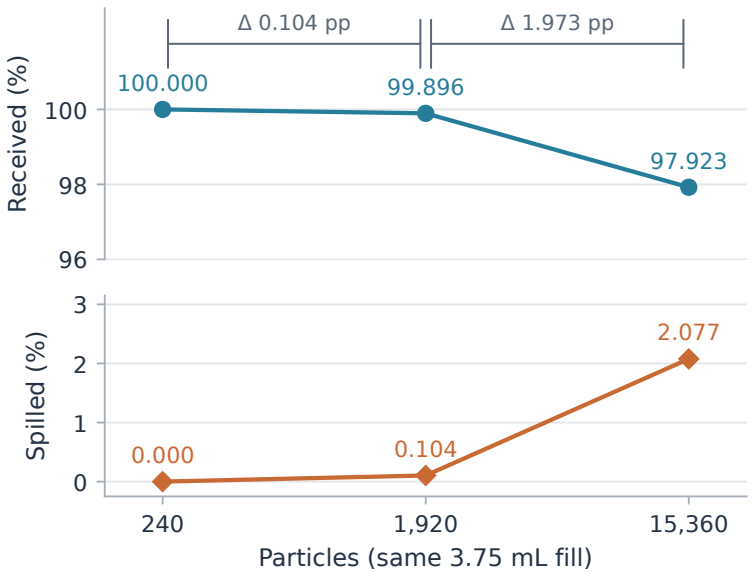
Native pouring: resolution and force-feedback checks

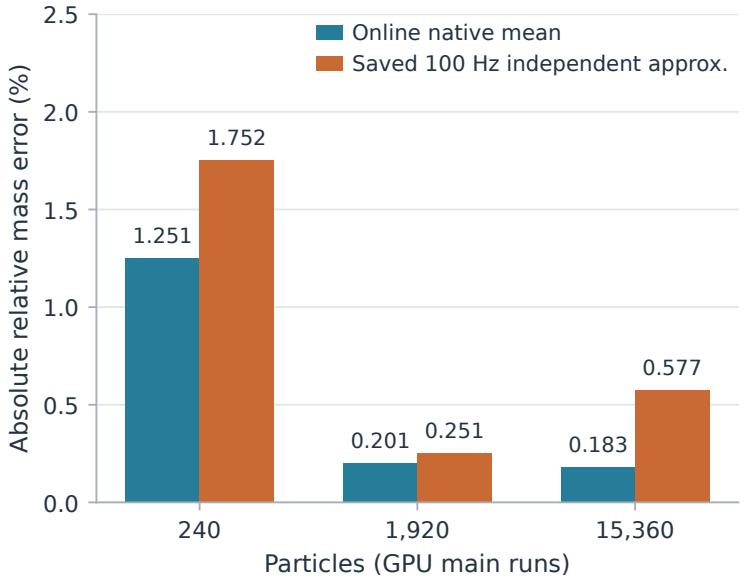
A01 local functional reconstruction | five full 23 s trajectories | single-run measurements; no error bars

(a) Final transfer across resolutions



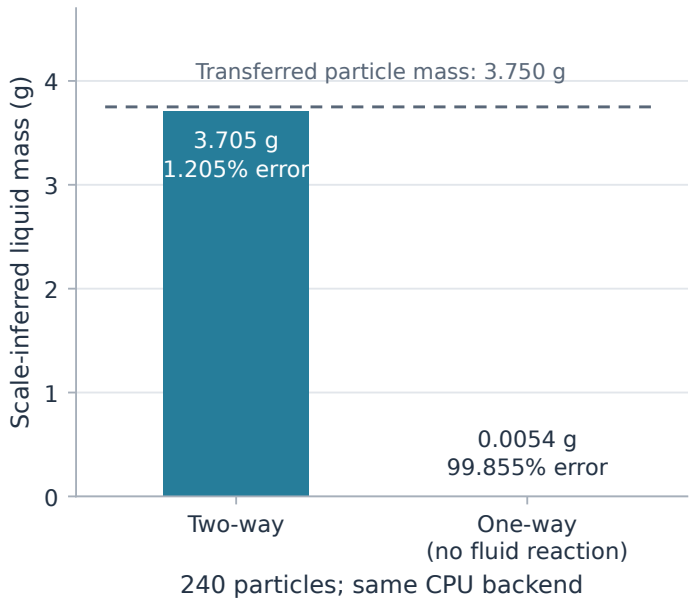
n = 1 run per resolution; separate y scales

(b) Main-run weighing error



Main-run exact substep CSVs not saved

(c) Matched CPU coupling control



240 particles; same CPU backend

Exact saved substeps; final 1 s mean

Each condition is one complete native 23 s run (n = 1); no uncertainty bars or statistical/asymptotic convergence claim. Adjacent final received/spilled differences are 0.104 and 1.973 percentage points.

GPU main-run scale readings use online native-substep means. Saved 100 Hz force traces provide an independent approximation only: exact GPU substep CSVs were not saved. CPU coupling controls retain exact substeps.

Retained failures: finer dam-break maximum error 14.52% exceeds the 10% gate; the raw particle pressure field fails the R^2/slope criteria. Mechanical wall traction is a separate observable.

Particle collision scope: tube/cup solids, bench and floor; excludes fluid–robot/rack. Soft rigid contact overlap and fine pregrasp peak spill 3.58% remain; final fine spill is 2.077%.

Independent Warp/MuJoCo reconstruction with outward-thickened walls and a scripted UR5e; not GRID or original hardware. Replay uses diagnostic particles rather than the original liquid rendering.

Sources: pour_runs/*/metrics.json and independent_review.json. Derived offline; accepted evidence unchanged.